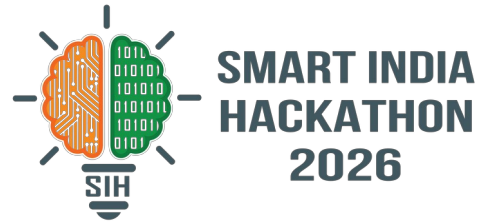
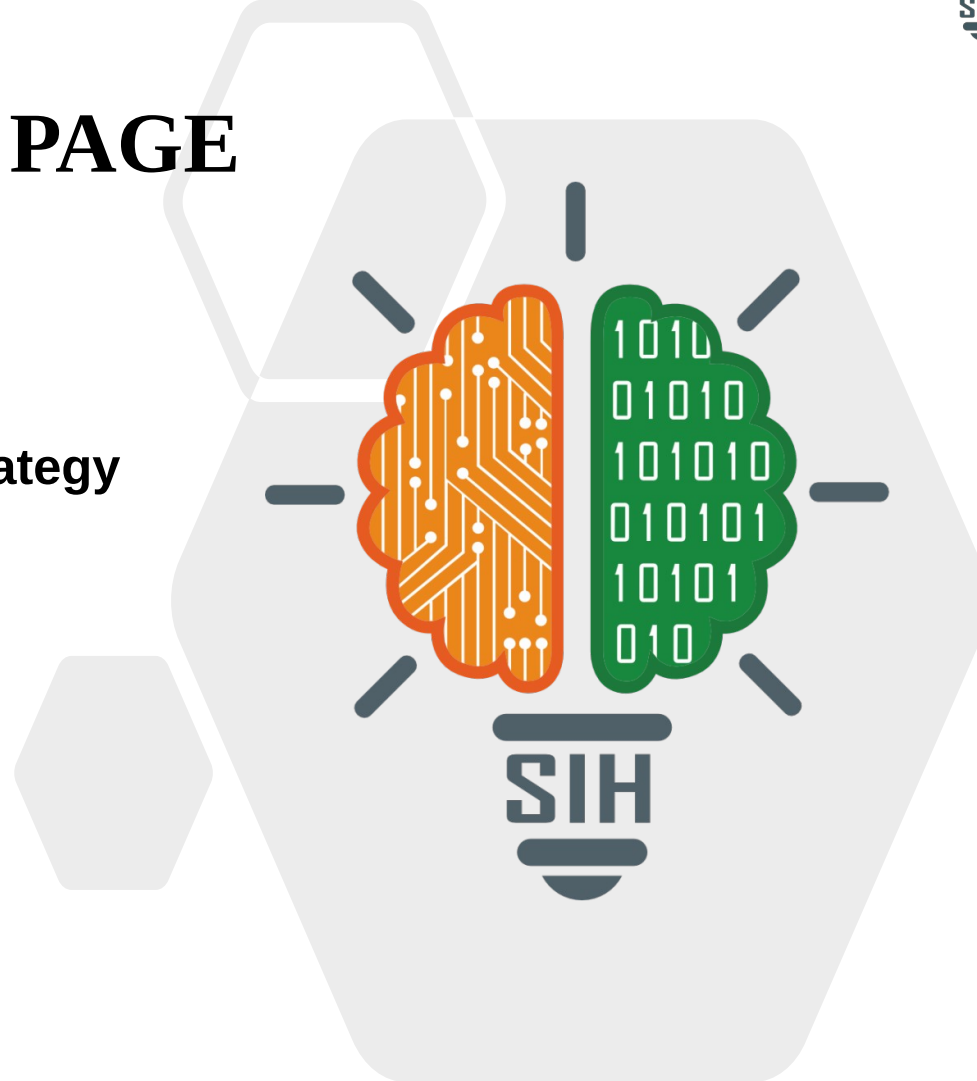


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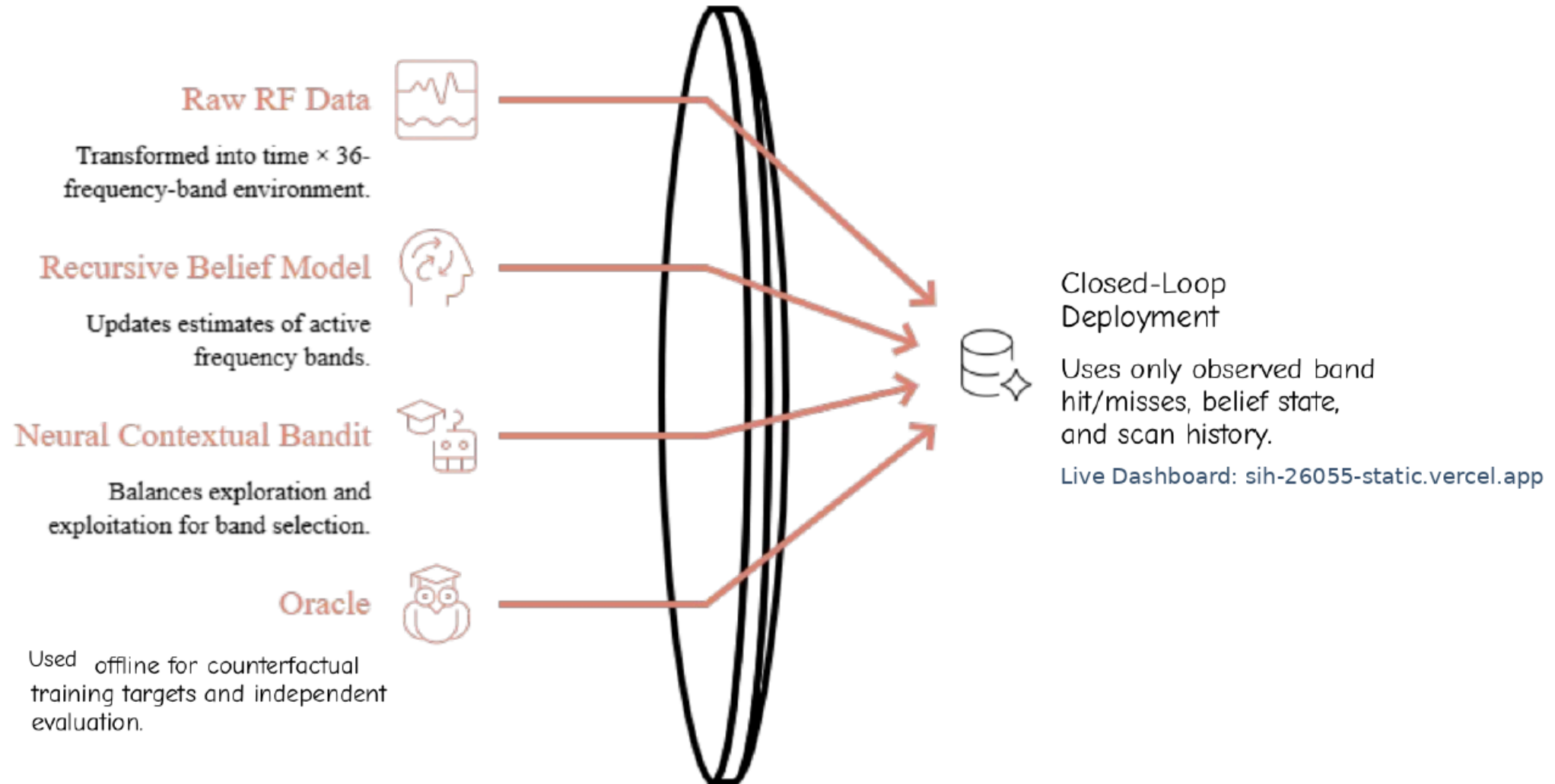


TITLE PAGE

- Problem Statement ID – SIH26055
- Problem Statement Title- Smart Scan Strategy for Electronic Warfare
- Theme- Robotics and Drones
- PS Category- Software
- Team ID-
- Team Name- CodeSutra



Adaptive RF Spectrum Scanning with Oracle-Guided Neural Bandits



Spectrum Scanning System Process



System
Architecture
Setup

Built using Python and PyTorch, utilizing a recursive belief updater and a temporal Neural Contextual Bandit with 3 Q ensemble heads and mean-Q action selection.



Spectrum
Scanning
Execution

The RF environment covers 0–18 GHz, divided into 36 frequency bands, with the bandit deciding which band to scan next.



Belief Model
Update

After every scan, the belief model updates its understanding of the spectrum using recent-scan history.



Oracle
Supervision
Training

The Oracle is used only offline to construct counterfactual training targets and evaluate the frozen policy.

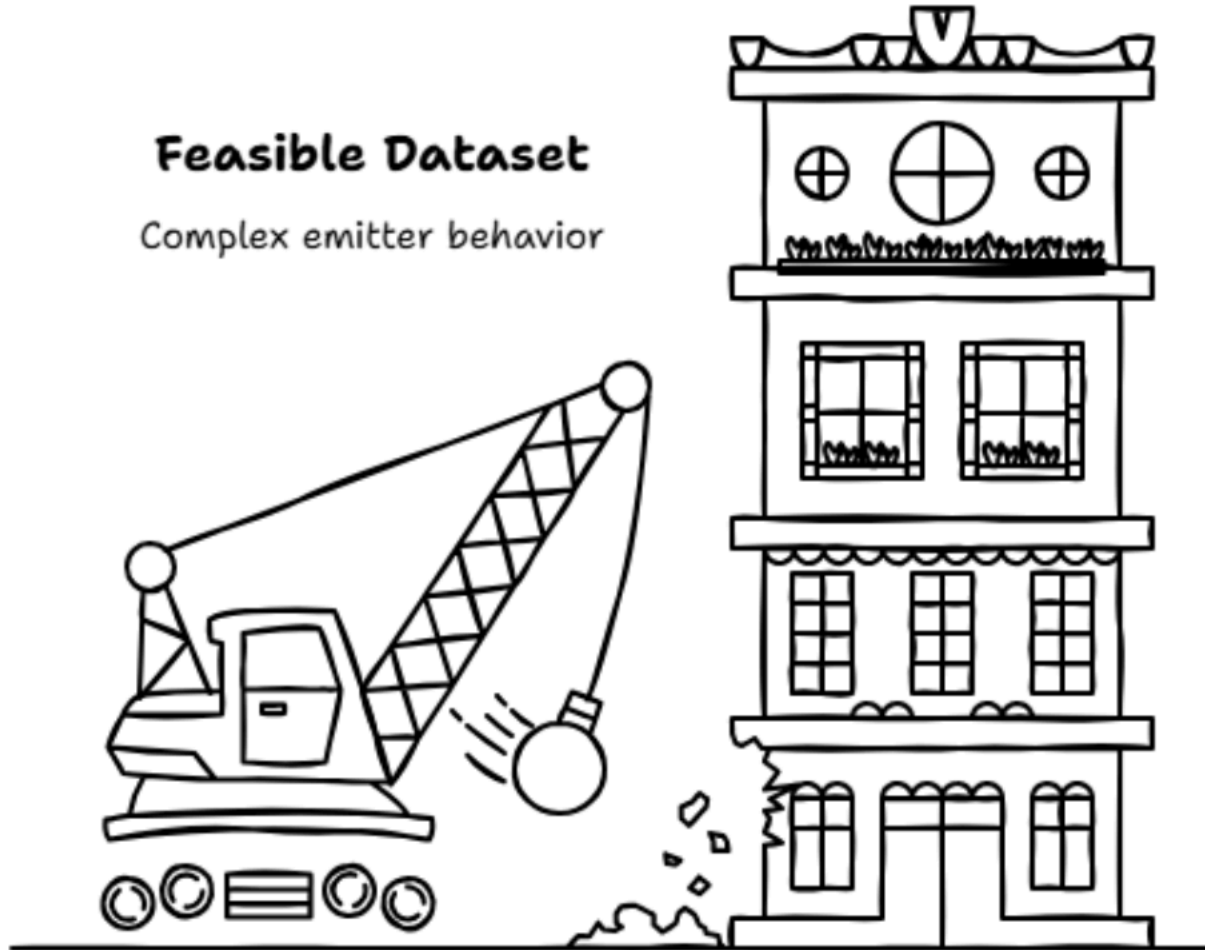
FEASIBILITY AND VIABILITY



Feasible Dataset Enables Bandit Training

Feasible Dataset

Complex emitter behavior



Temporal Memory

GRU-based temporal Q model

Q Selection

3-head Q ensemble +
mean-Q action selection

Risk Mitigation

Addresses partial observability

Simulation Training

Belief model and bandit trained
offline; frozen model at
deployment

IMPACT AND BENEFITS

System Impact and Benefits

No Prior Intelligence

Does not need prior intelligence about where or when emitters will transmit.

Superior Performance

Outperforms Random and Greedy-belief scanning in the held-out evaluation.

Faster Interception

Enables faster and smarter interception of transmissions across the full 0-18 GHz spectrum.

Adaptive

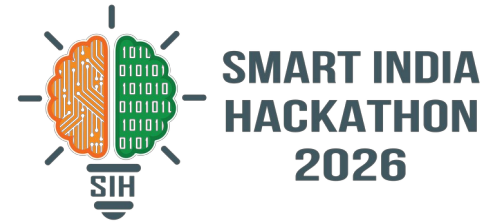
Adaptive Belief Updating

Updates band-activity beliefs from observed hits and misses during closed-loop scanning; model weights remain frozen at deployment.



System Capabilities

RESEARCH AND REFERENCES



- Neural Q ensembles for sequential decision-making under uncertainty
- GRU-based temporal Q models for partially observable spectrum-sensing environments
- Oracle-guided counterfactual offline reinforcement learning
- Electronic Warfare literature on frequency-agile emitter scanning and spectrum surveillance.